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# Hao Lu

PhD Candidate in Robotics

Google Scholar

School of Computing, Australian National University

Canberra, Australia

## Research Profile

Robotics researcher and systems engineer working on reliable robot planning and manipulation, with a focus on planning under uncertainty, sampling-based motion planning, manipulation and rearrangement, task and motion planning, combinatorial optimisation, and decision-making under partial observability. My doctoral research develops planning frameworks with theoretical guarantees while connecting algorithmic structure to practical robotic systems. My research builds on more than a decade in industrial systems engineering at Freescale/NXP (July 2013–February 2025), spanning embedded systems, software/hardware co-design, AI inference, performance optimisation, and customer-facing deployment.

## Research Interests

Motion planning; planning under uncertainty; manipulation and rearrangement; task and motion planning; kinodynamic planning; navigation; probabilistic robotics; partially observable decision-making; combinatorial optimisation; nonprehensile manipulation; robot learning and embodied AI; generative/world models for robotics; safety, reliability, and efficient decision-making.

## Education

**Australian National University (ANU)**, Canberra, Australia — Feb 2024 – Present
PhD Candidate in Computer Science (Robotics)
Supervisors: Dr Rahul Shome, Prof Hanna Kurniawati, Dr Marco Cheung

**Australian National University (ANU)**, Canberra, Australia — Feb 2021 – Dec 2023
Master of Machine Learning and Computer Vision | GPA: 6.92/7.0
Recipient of the ANU Postgraduate Medal for Academic Excellence

**University of Toronto**, Toronto, Canada — Sep 2022 – Apr 2023
Graduate Exchange Student, Department of Computer Science | GPA: 4.0/4.0

**Nankai University**, Tianjin, China — Sep 2010 – Jul 2013
Master of Engineering in Computer Science / Computer Architecture

## Peer-Reviewed Publications

**Contact-Rich Slide–Pick Manipulation Planning for Dense Cuboid Clutter with Parallel Grippers**.
**H. Lu**, R. Shome
*CoRL 2026 Workshop on Open Problems in Contact-Rich Loco-Manipulation. Accepted.* PDF

**Virtues of Ordered Chaos: Planning with Topple Actions in Tabletop Stack Rearrangement**.
**H. Lu**, R. Shome
*IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. Accepted.* PDF

**Optimal Knock-Pick Planning for Tightly Packed Tabletop Blocks With Parallel Grippers**.
**H. Lu**, R. Shome
*World Symposium on the Algorithmic Foundations of Robotics (WAFR), 2026. Accepted.* PDF

**Zero-Shot Generative World Priors for Planning and Object Reaching in Partially Observed Rooms**.
**H. Lu**, S. S. Bhattacharjee, D. Campbell, R. Shome
*RSS 2026 Workshop on Robot Planning in the Era of Foundation Models (FM4RoboPlan). Accepted.* PDF

**Sampling-based Motion Planning for Optimal Probability of Collision under Environment Uncertainty**.
**H. Lu**, H. Kurniawati, R. Shome
*IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024, pp. 3138–3145.* DOI

**Arbitor: A Numerically Accurate Hardware Emulation Tool for DNN Accelerators**.
C. Jiang, A. Jayarajan, **H. Lu**, G. Pekhimenko
*2023 USENIX Annual Technical Conference (USENIX ATC '23), pp. 519–536, 2023.*

## Research Experience and Projects

**PhD Researcher, Australian National University** Feb 2024 – Present

- Develop algorithms for reliable planning and decision-making under uncertainty, combining sampling-based planning, optimisation, probabilistic modelling, and robotic task/motion reasoning.
- Build end-to-end research systems using ROS2, MoveIt2, OMPL, Isaac Sim, collision checking, mathematical optimisation, and Monte Carlo evaluation; integrate algorithmic prototypes with simulated and real robotic platforms.

**Uncertainty-aware manipulation planning** *Oct 2025 – Present*

- Extending probability-of-collision-aware motion planning to manipulation problems with uncertain target-object and obstacle poses.
- Develop physics- and geometry-informed reasoning for grasping and manipulation under uncertainty, with the goal of coupling robust grasp/action feasibility with downstream motion-planning risk.
- Ongoing work targeted for submission to IEEE Robotics and Automation Letters (RA-L).

**Kinodynamic planning under environment and control uncertainty** *Aug 2026 – Present*

- Developing a sampling-based kinodynamic planning algorithm that extends the uncertainty-aware planning framework to systems without an exact steering function.
- Sample controls and durations for forward propagation; evaluate execution across uncertain environments and controls.
- Investigate asymptotic optimality using sparse kinodynamic exploration, scenario-based success probabilities, and pruning under uncertain dynamics.

**Sampled-World Contingent Task and Motion Planning with Model Repair** *Planned*

- Plan an online task and motion planning framework for unfamiliar environments with uncertain geometry and imperfect symbolic action models.
- Combine model-repair hypotheses with sampled geometric worlds; update scenario weights and states using execution observations.
- Couple a configuration-space roadmap with a belief graph to construct contingent policies that branch when observations distinguish scenarios.
- Investigate finite-horizon decisions, feasibility checking, and caching under bounded computation and execution risk.

**Optimal knock–pick planning for tightly packed tabletop blocks** *2025 – 2026*

- Formulated manipulation of tightly packed tabletop blocks using knock actions to expose graspable objects.
- Derived graph abstractions for polynomial-time optimal action selection under the studied grid assumptions.

**Planning with toppling actions for tabletop stack rearrangement** *Feb 2025 – 2026*

- Introduced purposeful toppling actions for rearranging dense tabletop stacks.
- Developed a graph-based toppling abstraction and a partially grounded task-and-motion planning framework for selecting pick/topple sequences under geometric constraints.

**Generative world priors for planning in partially observed rooms** *2025 – 2026*

- Investigated generative models as zero-shot priors over plausible unseen environment structure for downstream robot planning.
- Evaluated object-reaching decisions across multiple plausible worlds generated from partial observations.

**Motion planning for optimal probability of collision under environment uncertainty** *Feb 2024 – Oct 2024*

- Formulated sampling-based motion planning over environment samples to optimise probability of collision-free execution rather than path length alone.
- Used roadmap search together with mixed-integer optimisation to refine candidate paths and established asymptotic optimality properties for the probabilistic objective.
- Evaluated the framework on high-dimensional robotic manipulation problems.

**Arbitor: numerical emulation for DNN accelerators, University of Toronto** *Sep 2022 – Jan 2023*

- Co-developed a hardware emulation tool for rapidly evaluating DNN accelerator designs and the numerical effects of common DNN optimisations.
- Used machine-learning compiler infrastructure for accurate software emulation and fast prototyping.

**Selected earlier research/course projects** *2021 – 2023*

- Imitation learning combined with PILCO-style model-based policy learning for improved sample efficiency on higher-dimensional control tasks.
- Multi-object tracking evaluation comparing camera–LiDAR fusion and LiDAR-only baselines.
- Dictionary learning for text classification and source separation; FastICA; exponential-family PCA.
- Near-light photometric stereo and multimodal trajectory prediction for autonomous driving.

## Industry Experience

**Senior Principal Software System Engineer** Jul 2013 – Feb 2025
**NXP Semiconductors, Edge Processing Business Line**
*Freescale Semiconductor, Jul 2013 – Dec 2015; Freescale merged into NXP in 2015.*

- Progressed to Senior Principal Software System Engineer (September 2021), following promotions to Principal Software System Engineer (April 2018) and Senior Application Engineer (September 2015).
- Designed, developed, integrated, and deployed software components across embedded and edge-computing systems using C, C++, Python, Java, and architecture-specific assembly.
- Diagnosed software/hardware bottlenecks and optimised critical paths through algorithm redesign, compiler and instruction-level optimisation, cache/TLB analysis, multicore parallelism, accelerator offloading, and system-level profiling.
- Produced proof-of-concept systems, technical documentation, application notes, and engineering solutions; provided training and remote/on-site support to internal teams, partners, and customers.

**Selected Technical Projects**

- **Machine Learning Foundation Project:** collaborated with NXP's CTO organisation on internal infrastructure for creating, adapting, deploying, optimising, and evaluating deep-learning/foundation models for edge-AI applications.
- **NPU and Heterogeneous AI Inference Platform:** developed models and deployment workflows for NXP's Neutron AI inference engine; investigated heterogeneous execution across NPU, GPU, ARM Cortex-A, and ARM Cortex-M backends and optimisation of drivers, runtimes, firmware, and supporting software stacks.
- **i.MX 9 Smart Meter / Edge-AI Project:** developed and deployed AI algorithms on i.MX 9 NPU platforms for appliance anomaly detection, diagnosis/prediction, and non-intrusive load/energy disaggregation.
- **5G/O-RAN Systems:** contributed to 5G RU/DU, CPE, small-cell, ISC, femtocell, and O-RAN solutions across LA12xx/LS104x/LX2160-class platforms, integrating and supporting L1/L2/L3 and accelerator-enabled system components.
- **System Performance Analysis and Optimisation:** used tools such as CodeWarrior and OProfile together with compiler, architecture, cache/TLB, bus, accelerator, and multicore analysis to improve end-to-end customer software performance.

## Technical Skills

**Programming:** C++, Python, C, Java, MATLAB; architecture- and accelerator-specific assembly
**Robotics and Planning:** ROS2, MoveIt2, OMPL; motion planning, task and motion planning, kinodynamic planning, probabilistic planning, collision checking
**Optimisation and Algorithms:** MILP/MIQCP, Gurobi, combinatorial optimisation, graph algorithms, randomized algorithms, approximation algorithms, dynamic programming, Monte Carlo methods
**Machine Learning / AI:** PyTorch, TensorFlow, probabilistic modelling, generative models, model deployment and inference optimisation, edge AI, heterogeneous computing
**Simulation and Systems:** Isaac Sim, Gazebo, Linux/Ubuntu, Git, CMake/Make, GCC/LLVM, cross-compilation; ARM and PowerPC architectures, SIMD/NEON
**Libraries / Tools:** OpenCV, Boost, Eigen; profiling and low-level system performance analysis

## Teaching Experience

**COMP3600 – Algorithms, Australian National University** Jul 2026 – Present
Tutor for algorithm design and analysis, including trees, balanced search structures, randomized algorithms, graph algorithms, dynamic programming, and complexity analysis; prepare tutorial explanations, solutions, and marking guidance.

**COMP8620 – Advanced Topics in Artificial Intelligence, ANU** Jul 2026 – Present
Tutor for advanced AI and robotics topics including configuration spaces, motion planning, MDPs, POMDPs, belief-state planning, value/policy iteration, RTDP, and reinforcement learning.

**COMP8600 – Statistical Machine Learning, ANU** Feb 2026 – Jun 2026
Tutor for probabilistic modelling, Bayesian inference, kernel methods, Gaussian processes, optimisation, and statistical learning.

**COMP8460 – Advanced Algorithms, ANU** Feb 2026 – Jun 2026
Tutor for advanced algorithm design, randomized and approximation algorithms, graph algorithms, and complexity analysis.

**COMP8620 – Advanced Topics in Artificial Intelligence, ANU** Jul 2025 – Nov 2025
Delivered tutorials on motion planning, Monte Carlo methods, Bayesian inference, MDPs, POMDPs, and reinforcement learning; supported demonstrations, Q&A, and assessment.

**COMP6670 – Introduction to Machine Learning, ANU** Jul 2025 – Nov 2025
Tutored mathematical foundations of machine learning, including linear algebra, probability, vector calculus, optimisation, and classical ML algorithms.

## Awards and Recognition

**Academic**

- ANU PhD Scholarship and ANU HDR Fee Remission Merit Scholarship, 2024.
- ANU Postgraduate Medal for Academic Excellence, 2023.

**Industry**

- NXP Top Achiever, 2015–2022.
- NXP China R&D Innovation Award, 2023.
- NXP Bravo Award, 2021.
- NXP Bravo Award, 2018.
- Freescale Rockstar Award, 2015.